# Similar Predictions, Different Weights: How AdamW and Muon Train a Small Language Model

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Juntang Wang and Víctor Conchello · November 1, 2026

**Summary.** We train a small language model with AdamW and with Muon, from one initialization and on the same mini-batches. (1) Muon trains faster, on a path that differs from AdamW's less than a change of data order does. (2) Muon's weight and update matrices have more evenly sized singular values than AdamW's, beyond data-order variation; no prediction statistic differs. (3) A run that switches optimizer mid-training ends with predictions as close to the new optimizer's run as two data orders are. Its updates change at once, but the stable rank of its weights and its MLP sparsity keep much of their old values, more so the later the switch.

**Setup.** *Model.* We train a 14M-parameter nanoGPT (Karpathy, 2022) on TinyStories (Eldan and Li, 2023) for 10,000 steps of 128 sequences of 256 tokens (Appendix A lists the model, data, training and optimizer settings).

*Optimizers.* Muon (Jordan et al., 2024) orthogonalizes the update of each weight matrix in the transformer blocks: it sets the update's singular values to about 1. AdamW trains Muon's other parameters. Each optimizer uses the best of five learning rates. All runs start from the same weights. Each optimizer runs on the same five *data orders*.

**Step 1: how the trajectories differ (Fig. 1).** *Comparing runs.* The distance between two runs is the mean Jensen–Shannon divergence (JS, in bits) of their next-token predictions on held-out text. We compare runs at equal step and at equal training loss; at equal loss, speed does not count as a difference. The *floor* is the mean distance between two data orders of one optimizer. A distance of one floor means "as different as two data orders".

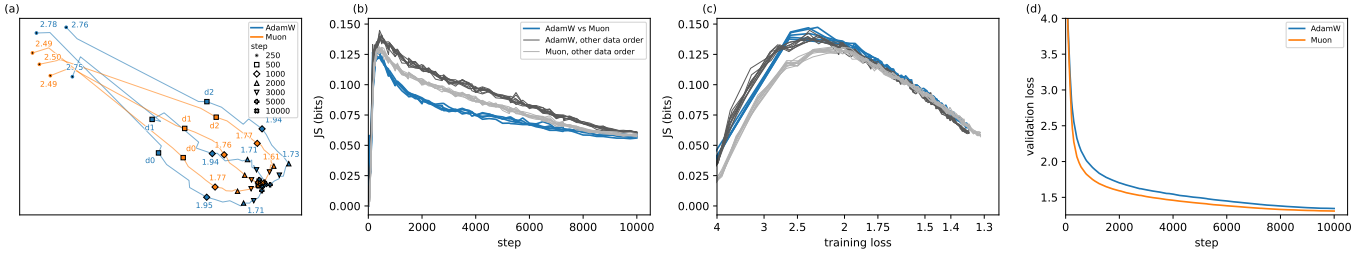



**Figure 1:** AdamW and Muon. (a) Checkpoints from step 250 on, mapped to 2-D by densMAP (shows neighbours, not distances); labels: training loss and data order. (b) JS between runs at equal step. (c) The same at equal training loss. (d) Validation loss.

*Faster, and on a slightly different path.* Muon reaches AdamW's final validation loss, 1.344, by step 7500 (d). Since they share the initialization and batches, if Muon just followed AdamW's path faster, they would be closer at equal loss (c) than at equal step (b). The opposite happens: from step 250 on, their distance is larger at equal loss than at equal step (at step 1000, 0.131 against 0.102). So the paths differ, but less than a change of data order does: from step 500 on, at equal step, AdamW and Muon on the same data order are closer than two data orders of one optimizer (a, b).

*Endpoints.* The final AdamW and Muon models are about as far apart as two runs of one optimizer on different data orders (b), but Muon ends at a lower validation loss: 1.309 against 1.344 (d).

**Step 2: what differs along the way.** We measure 30 properties of each model at 34 points of training, in five groups: weights $W$, one-step updates $\Delta W$, loss curvature (Hessian $H$), internal activations and predictions (Appendix B defines each metric and compares all of them in Tables 3 and 4). Table 4 also gives the values for SGD with momentum.

We compare the optimizers by each metric's ratio AdamW/Muon at training loss 1.81, on three data orders each (Fig. 2, Table 3). The ± gives its range over data orders; if it excludes 1, the optimizers differ more than data orders do.

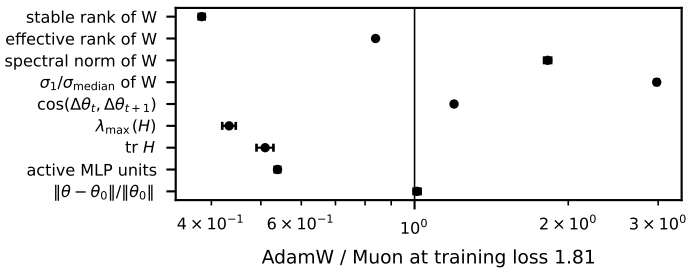



**Figure 2:** AdamW's value divided by Muon's at training loss 1.81, for selected metrics; three data orders each; bars: range across data orders.

- *Weights $W$.* Muon's weight matrices have more evenly sized singular values. Their stable rank $\|W\|_F^2/\|W\|_2^2$ is 86 against AdamW's 33 (Muon/AdamW $2.61 \pm 0.04$), and their effective rank 325 against 272. Their spectral norm (2.9 against 5.4) and largest-to-median singular-value ratio $\sigma_1/\sigma_{\text{median}}$ (2.7 against 8.0) are lower. Both move similarly far from the initialization (ratio $1.01 \pm 0.02$): they differ in the shape of their weights, not in how far they travel.
- *One-step updates $\Delta W$.* Consecutive Muon steps are less alike (cosine 0.74 against 0.88). Their stable rank is much higher (220 against 17; Table 3), but by construction: Muon orthogonalizes every update.
- *Loss curvature (Hessian $H$).* Muon is about twice as sharp: the top Hessian eigenvalue $\lambda_{\max}$ is 14 against 6.2, the trace $\operatorname{tr} H$ 2770 against 1400. This holds even though both take steps of similar relative size ($\|\Delta\theta\|/\|\theta\|$ is $6.53 \cdot 10^{-3}$ for Muon, $6.56 \cdot 10^{-3}$ for AdamW), so it is not an artifact of their different learning rates.
- *Internal activations.* AdamW's MLP is sparser: 12% of its units are active, against 22% for Muon.
- *Predictions.* No prediction metric separates the two: every AdamW/Muon range contains 1 (Table 3).

**Step 3: swaps (Fig. 3).** *Swaps.* A *swap* switches a run of one optimizer (the *parent*, on data order 0) to the other optimizer at a *fork step*, then trains it to step 10,000 on the parent's batches (Appendix C gives the fork steps, the optimizer state at the swap and the measurement steps). An optimizer's *own run* is its run from the start. We swap at nine fork steps from 50 to 4000, in both directions. We give each metric's value at step 10,000 as a fraction of the way from the parent's value (0) to the own run's (1).

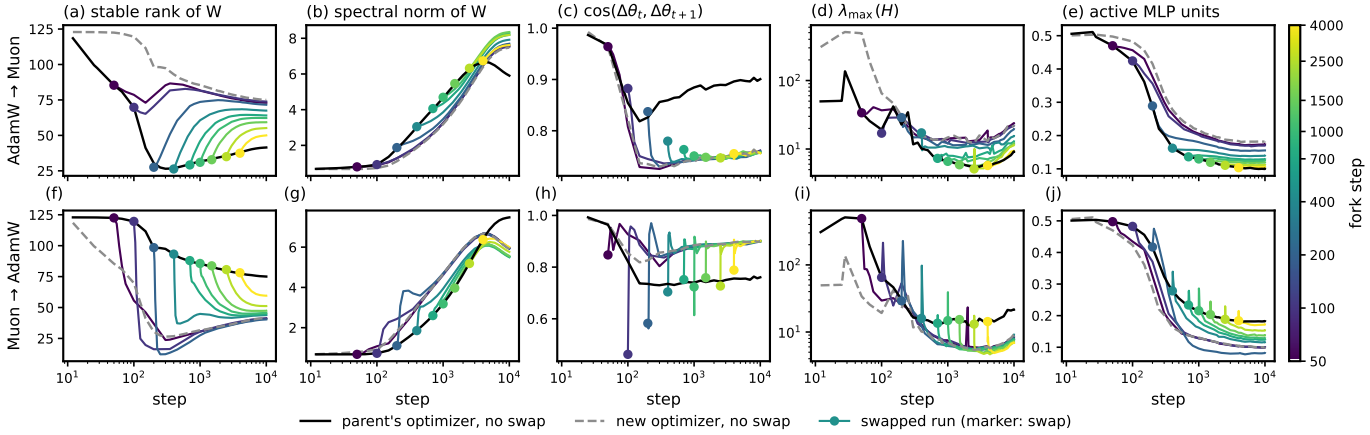



**Figure 3:** Swaps on data order 0: five metrics against the step. Top: AdamW to Muon; bottom: Muon to AdamW. Black: the parent's optimizer without a swap; grey dashed: the new optimizer's own run; colour: swapped runs, by fork step (dot: the swap).

*Weights $W$.* The spectral norm ends near or beyond the own run's (0.94–1.56 of the way; b, g). The stable rank moves less the later the swap: from fork step 4000, 0.25 (AdamW to Muon) and 0.46 (Muon to AdamW) of the way (a, f).

*One-step updates $\Delta W$.* The update cosine switches at once: within 50 steps of the swap, the cosine between consecutive updates moves to the new optimizer's level (c, h). At step 10,000 it is 0.99–1.03 of the way.

*Loss curvature (Hessian $H$).* $\lambda_{\max}$ differs by direction (d, i): after a switch to AdamW, it falls to AdamW's level (1.00–1.18 of the way), often after a brief spike; after a switch to Muon, it reaches Muon's level only for swaps up to step 100, and from step 700 on at most 0.30 of the way. Part of this may reflect the swap's change of learning rate.

*Internal activations.* The fraction of active MLP units moves less the later the swap: from fork step 4000, only 0.07 (AdamW to Muon) and 0.13 (Muon to AdamW) of the way (e, j).

*Predictions.* Every swapped run ends as close to the new optimizer's own run as two data orders are (0.86–0.95 floors).

**Next steps.** (i) Repeat Steps 1–3 on a larger model (TinyStories-30M) and on character-level Shakespeare. (ii) For each property that differs in Step 2: does it follow directly from Muon's orthogonalized updates, or does it emerge during training? (iii) Do these differences explain Step 1, Muon's faster training and different path? Is any of them a reason Muon does better? (iv) Talk (Dec 1–4); paper, code, Colab (Dec 5–11). https://github.com/qqgjyx/MIT-6.7910

## References

Eldan, R. and Y. Li (2023). *TinyStories: How Small Can Language Models Be and Still Speak Coherent English?*

Jordan, K., Y. Jin, V. Boza, et al. (2024). *Muon: An Optimizer for Hidden Layers in Neural Networks*. Blog post. URL: https://kellerjordan.github.io/posts/muon/.

Karpathy, A. (2022). *nanoGPT*. Software. URL: https://github.com/karpathy/nanoGPT.

# A Hyperparameters

**Table 1:** Hyperparameters shared by the AdamW and Muon runs.

| | |
|---|---|
| Runs | AdamW and Muon × data orders d0–d4 (10) |
| Model | nanoGPT, pre-LayerNorm blocks |
| Layers, heads, width | 6, 6, 384 |
| MLP | GELU, width 1536 |
| Biases, dropout | none, none |
| Output head | untied |
| Attention projections | separate query, key, value |
| Context | 256 tokens |
| Parameters | 13.9M (10.6M without token and position embeddings and head) |
| Data | TinyStories (GPT-4 version) |
| Tokenizer | byte-level BPE, 4096 tokens, trained on the training split |
| Batch | 128 windows × 256 tokens |
| Window offsets | from the training split; generator seeded with (data order, step), same for every optimizer |
| Steps | 10,000 |
| Tokens | 328M (about 0.6 epochs) |
| Learning rate | $\eta\,\varphi(t)$ at step $t$, every parameter group |
| Warmup | linear, $\varphi(t) = (t+1)/200$ for $t < 200$ |
| Decay | cosine to $\varphi = 0.1$ at step 10,000 |
| Gradient clipping | none |
| Initialization seed | 0, all runs |
| Training precision | bfloat16 autocast |
| Prediction, Hessian precision | float32, no autocast (TF32 matrix products allowed) |
| Logged predictions | next-token distributions: 128 fixed validation sequences × 16 evenly spaced positions (2,048), as float16 log-probabilities |
| Training loss | 64 fixed training sequences, all positions; used to match runs at equal loss |
| Floor | mean JS between two data orders of one optimizer, at equal training loss; for two optimizers, the larger of their floors |
| Both logged at | every 250th step + 30 log-spaced steps in $[1, 10{,}000]$ |
| Hessian | top eigenvalue, 20 Lanczos iterations, 32 fixed training sequences; AdamW also of $P^{-1/2}HP^{-1/2}$, with $P$ AdamW's diagonal preconditioner |
| Measured at | every 500th step + 30 log-spaced steps |

**Table 2:** Optimizer-specific hyperparameters. $n_{\text{out}}$, $n_{\text{in}}$: output and input size of the matrix.

| | AdamW | Muon |
|---|---|---|
| $\eta$ | $2 \cdot 10^{-3}$ | 0.01 |
| $\eta$ choice | best final validation loss of a 5-rate sweep on d0 | as AdamW |
| Update | bias-corrected moments | Nesterov momentum; 5 quintic Newton–Schulz steps in bfloat16 |
| Momentum | $\beta = (0.9, 0.95)$ | $\beta = 0.95$ |
| $\epsilon$ | $10^{-8}$ | – |
| Update scale | – | $\sqrt{\max(1, n_{\text{out}}/n_{\text{in}})}$ |
| Weight decay | 0.1, decoupled; none on LayerNorm gains | none |
| Parameters | all | query, key, value, attention-output and both MLP matrices of every block |
| Aux. AdamW | – | embeddings, head, LayerNorm gains |
| Aux. $\eta$, $\beta$, $\epsilon$ | – | $2 \cdot 10^{-3}$, $(0.9, 0.95)$, $10^{-8}$ |
| Aux. weight decay | – | none |

# B Metrics

The Step-2 runs save checkpoints every 1000 steps, at 12 log-spaced steps, and at three consecutive steps $t, t+1, t+2$ for 24 values of $t$ from 25 to 9998 (to measure one-step updates); we measure the metrics at 34 of these per run, and the Step-3 swaps at the steps in Table 5. *Weights*: for all parameters together, $\|\theta\|_2$, $\|\theta - \theta_0\|/\|\theta_0\|$ and $\|\theta\|_\infty/\|\theta\|_2$; and, averaged over the 36 weight matrices $W$ of the blocks, the stable rank $\|W\|_F^2/\|W\|_2^2$, the effective rank $\exp H(\sigma/\sum\sigma)$ with $H$ the entropy of the normalized singular values $\sigma$, the spectral norm, and $\sigma_1/\sigma_{\text{median}}$. *Updates* $\Delta\theta_t = \theta_{t+1} - \theta_t$: $\|\Delta\theta_t\|/\|\theta_t\|$, its cosine with $\Delta\theta_{t+1}$ and with the negative gradient $-g_t$ on step $t$'s batch, and the stable and effective

rank of its block matrices $\Delta W$. *Curvature* of the training loss on 32 fixed training sequences, with Hessian $H$: $\lambda_{\max}(H)$ (20 Lanczos iterations, logged during training), $\operatorname{tr} H$ (10 Rademacher probes), $\operatorname{tr} H/\lambda_{\max}$, and the batch sharpness $\mathbb{E}_B[g_B^\top H_B g_B]/\mathbb{E}_B\|g_B\|^2$, where $g_B$ and $H_B$ are the gradient and Hessian on a mini-batch $B$. *Predictions*, on the 2,048 held-out positions: entropy, mean top-1 probability, calibration error (15 bins), top-1 accuracy, loss on the targets in the top and the bottom third of training-set frequency and their difference, and JS to the unigram and to the bigram model of the training split. *Activations*, on 32 held-out sequences: attention entropy and attention on the first token (means over layers, heads and positions), fraction of MLP pre-activations above 0, RMS of the residual stream after the last block, and effective rank of the centred final features.

SGD, the reference in Table 4, uses heavy-ball momentum 0.9 and learning rate 0.06, the only rate tried between 0.03 and 0.12 that did not diverge on any of the three data orders; everything else is as in Table 1.

**Table 3:** All metrics of Step 2 at training loss 1.81: mean ± half-range over data orders 0–2, and the ratio AdamW/Muon. If AdamW's values range over $[a_{\min}, a_{\max}]$ and Muon's over $[b_{\min}, b_{\max}]$, the ratio ranges over $[a_{\min}/b_{\max},\ a_{\max}/b_{\min}]$, given as midpoint ± half-width. Rows marked ($\times 10^k$) are in units of $10^k$. Bold: the five ratios farthest from 1 whose range excludes 1.

| | AdamW | Muon | A/M |
|---|---|---|---|
| *Weights $W$* | | | |
| weight norm $\|\theta\|_2$ | $224 \pm 2$ | $224 \pm 0$ | $0.997 \pm 0.013$ |
| distance from init. $/\|\theta_0\|$ | $2.18 \pm 0.03$ | $2.15 \pm 0.01$ | $1.01 \pm 0.02$ |
| largest entry $/\|\theta\|_2$ ($\times 10^{-3}$) | $7.04 \pm 0.06$ | $6.45 \pm 0.02$ | $1.09 \pm 0.01$ |
| stable rank of $W$ | $33.0 \pm 0.3$ | $86.2 \pm 0.3$ | $\mathbf{0.383 \pm 0.005}$ |
| effective rank of $W$ | $272 \pm 0$ | $325 \pm 0$ | $0.839 \pm 0.000$ |
| spectral norm of $W$ | $5.39 \pm 0.07$ | $2.95 \pm 0.01$ | $1.82 \pm 0.03$ |
| $\sigma_1/\sigma_{\text{median}}$ of $W$ | $8.01 \pm 0.05$ | $2.68 \pm 0.00$ | $\mathbf{2.98 \pm 0.03}$ |
| *One-step updates $\Delta W$* | | | |
| step size $\|\Delta\theta\|/\|\theta\|$ ($\times 10^{-3}$) | $6.56 \pm 0.08$ | $6.53 \pm 0.03$ | $1.01 \pm 0.02$ |
| stable rank of $\Delta W$ | $16.7 \pm 0.3$ | $220 \pm 0$ | $\mathbf{(7.66 \pm 0.14) \times 10^{-2}}$ |
| effective rank of $\Delta W$ | $228 \pm 1$ | $369 \pm 0$ | $0.618 \pm 0.003$ |
| cosine of consecutive steps | $0.883 \pm 0.002$ | $0.738 \pm 0.003$ | $1.19 \pm 0.01$ |
| cosine of step and $-g$ | $0.244 \pm 0.004$ | $0.217 \pm 0.007$ | $1.12 \pm 0.06$ |
| *Loss curvature (Hessian $H$)* | | | |
| top Hessian eigenvalue | $6.18 \pm 0.10$ | $14.3 \pm 0.2$ | $\mathbf{0.433 \pm 0.013}$ |
| batch sharpness | $0.974 \pm 0.254$ | $3.13 \pm 0.37$ | $\mathbf{0.332 \pm 0.116}$ |
| Hessian trace | $1400 \pm 40$ | $2770 \pm 20$ | $0.510 \pm 0.019$ |
| trace / top eigenvalue | $224 \pm 6$ | $196 \pm 7$ | $1.16 \pm 0.08$ |
| *Internal activations* | | | |
| attention entropy (nats) | $2.51 \pm 0.03$ | $2.79 \pm 0.02$ | $0.899 \pm 0.014$ |
| attention on first token ($\times 10^{-2}$) | $1.75 \pm 0.11$ | $1.98 \pm 0.05$ | $0.908 \pm 0.079$ |
| active MLP units | $0.120 \pm 0.001$ | $0.222 \pm 0.001$ | $0.539 \pm 0.007$ |
| residual-stream RMS | $4.73 \pm 0.05$ | $3.38 \pm 0.02$ | $1.40 \pm 0.03$ |
| effective rank of features | $312 \pm 2$ | $320 \pm 0$ | $0.977 \pm 0.007$ |
| *Predictions* | | | |
| prediction entropy (nats) | $1.89 \pm 0.01$ | $1.88 \pm 0.01$ | $1.01 \pm 0.01$ |
| mean top-1 probability | $0.561 \pm 0.003$ | $0.563 \pm 0.001$ | $0.994 \pm 0.007$ |
| calibration error ($\times 10^{-2}$) | $3.25 \pm 0.17$ | $3.05 \pm 0.46$ | $1.09 \pm 0.22$ |
| top-1 accuracy | $0.546 \pm 0.005$ | $0.544 \pm 0.005$ | $1.01 \pm 0.02$ |
| loss on frequent targets | $0.896 \pm 0.004$ | $0.892 \pm 0.008$ | $1.00 \pm 0.01$ |
| loss on rare targets | $3.10 \pm 0.02$ | $3.10 \pm 0.01$ | $1.00 \pm 0.01$ |
| rare minus frequent loss | $2.21 \pm 0.02$ | $2.21 \pm 0.01$ | $0.996 \pm 0.015$ |
| JS to unigram model | $0.800 \pm 0.001$ | $0.802 \pm 0.001$ | $0.998 \pm 0.002$ |
| JS to bigram model | $0.469 \pm 0.001$ | $0.471 \pm 0.001$ | $0.996 \pm 0.004$ |

**Table 4:** The metrics of Table 3 with SGD as a reference: ratios AdamW/SGD and Muon/SGD. Bold: in each ratio column, the five ratios farthest from 1 whose range excludes 1.

| | AdamW | Muon | SGD | A/S | M/S |
|---|---|---|---|---|---|
| *Weights $W$* | | | | | |
| weight norm $\|\theta\|_2$ | $224 \pm 2$ | $224 \pm 0$ | $110 \pm 0$ | $2.04 \pm 0.02$ | $2.05 \pm 0.01$ |
| distance from init. $/\|\theta_0\|$ | $2.18 \pm 0.03$ | $2.15 \pm 0.01$ | $0.486 \pm 0.006$ | $4.47 \pm 0.11$ | $4.42 \pm 0.07$ |
| largest entry $/\|\theta\|_2$ ($\times 10^{-3}$) | $7.04 \pm 0.06$ | $6.45 \pm 0.02$ | $39.5 \pm 1.0$ | $0.180 \pm 0.006$ | $\mathbf{0.165 \pm 0.005}$ |
| stable rank of $W$ | $33.0 \pm 0.3$ | $86.2 \pm 0.3$ | $56.2 \pm 5.6$ | $0.614 \pm 0.069$ | $1.60 \pm 0.17$ |
| effective rank of $W$ | $272 \pm 0$ | $325 \pm 0$ | $321 \pm 1$ | $0.849 \pm 0.003$ | $1.01 \pm 0.00$ |
| spectral norm of $W$ | $5.39 \pm 0.07$ | $2.95 \pm 0.01$ | $1.47 \pm 0.11$ | $3.61 \pm 0.31$ | $1.98 \pm 0.15$ |
| $\sigma_1/\sigma_{\text{median}}$ of $W$ | $8.01 \pm 0.05$ | $2.68 \pm 0.00$ | $4.37 \pm 0.55$ | $1.79 \pm 0.23$ | $0.599 \pm 0.074$ |
| *One-step updates $\Delta W$* | | | | | |
| step size $\|\Delta\theta\|/\|\theta\|$ ($\times 10^{-3}$) | $6.56 \pm 0.08$ | $6.53 \pm 0.03$ | $0.624 \pm 0.047$ | $\mathbf{10.9 \pm 1.0}$ | $\mathbf{10.8 \pm 0.9}$ |
| stable rank of $\Delta W$ | $16.7 \pm 0.3$ | $220 \pm 0$ | $8.37 \pm 0.48$ | $2.04 \pm 0.15$ | $\mathbf{26.6 \pm 1.6}$ |
| effective rank of $\Delta W$ | $228 \pm 1$ | $369 \pm 0$ | $205 \pm 2$ | $1.12 \pm 0.02$ | $1.81 \pm 0.02$ |
| cosine of consecutive steps | $0.883 \pm 0.002$ | $0.738 \pm 0.003$ | $0.857 \pm 0.004$ | $1.03 \pm 0.01$ | $0.863 \pm 0.007$ |
| cosine of step and $-g$ | $0.244 \pm 0.004$ | $0.217 \pm 0.007$ | $0.447 \pm 0.010$ | $0.550 \pm 0.022$ | $0.491 \pm 0.027$ |
| *Loss curvature (Hessian $H$)* | | | | | |
| top Hessian eigenvalue | $6.18 \pm 0.10$ | $14.3 \pm 0.2$ | $50.2 \pm 0.6$ | $\mathbf{0.123 \pm 0.004}$ | $0.284 \pm 0.008$ |
| batch sharpness | $0.974 \pm 0.254$ | $3.13 \pm 0.37$ | $6.38 \pm 0.50$ | $\mathbf{0.162 \pm 0.051}$ | $0.504 \pm 0.095$ |
| Hessian trace | $1400 \pm 40$ | $2770 \pm 20$ | $(2.55 \pm 0.04) \times 10^4$ | $\mathbf{(5.55 \pm 0.25) \times 10^{-2}}$ | $\mathbf{0.109 \pm 0.002}$ |
| trace / top eigenvalue | $224 \pm 6$ | $196 \pm 7$ | $507 \pm 9$ | $0.447 \pm 0.020$ | $0.385 \pm 0.021$ |
| *Internal activations* | | | | | |
| attention entropy (nats) | $2.51 \pm 0.03$ | $2.79 \pm 0.02$ | $3.84 \pm 0.01$ | $0.655 \pm 0.009$ | $0.728 \pm 0.007$ |
| attention on first token ($\times 10^{-2}$) | $1.75 \pm 0.11$ | $1.98 \pm 0.05$ | $1.82 \pm 0.07$ | $0.972 \pm 0.095$ | $1.07 \pm 0.07$ |
| active MLP units | $0.120 \pm 0.001$ | $0.222 \pm 0.001$ | $0.262 \pm 0.002$ | $0.458 \pm 0.008$ | $0.850 \pm 0.012$ |
| residual-stream RMS | $4.73 \pm 0.05$ | $3.38 \pm 0.02$ | $0.244 \pm 0.003$ | $\mathbf{19.4 \pm 0.5}$ | $\mathbf{13.8 \pm 0.3}$ |
| effective rank of features | $312 \pm 2$ | $320 \pm 0$ | $297 \pm 2$ | $1.05 \pm 0.01$ | $1.07 \pm 0.01$ |
| *Predictions* | | | | | |
| prediction entropy (nats) | $1.89 \pm 0.01$ | $1.88 \pm 0.01$ | $1.94 \pm 0.00$ | $0.971 \pm 0.006$ | $0.966 \pm 0.007$ |
| mean top-1 probability | $0.561 \pm 0.003$ | $0.563 \pm 0.001$ | $0.558 \pm 0.001$ | $1.00 \pm 0.01$ | $1.01 \pm 0.00$ |
| calibration error ($\times 10^{-2}$) | $3.25 \pm 0.17$ | $3.05 \pm 0.46$ | $2.97 \pm 0.28$ | $1.08 \pm 0.16$ | $1.02 \pm 0.25$ |
| top-1 accuracy | $0.546 \pm 0.005$ | $0.544 \pm 0.005$ | $0.554 \pm 0.003$ | $0.987 \pm 0.013$ | $0.980 \pm 0.014$ |
| loss on frequent targets | $0.896 \pm 0.004$ | $0.892 \pm 0.008$ | $0.851 \pm 0.012$ | $1.05 \pm 0.02$ | $1.05 \pm 0.02$ |
| loss on rare targets | $3.10 \pm 0.02$ | $3.10 \pm 0.01$ | $3.11 \pm 0.04$ | $0.999 \pm 0.018$ | $0.999 \pm 0.015$ |
| rare minus frequent loss | $2.21 \pm 0.02$ | $2.21 \pm 0.01$ | $2.26 \pm 0.03$ | $0.979 \pm 0.020$ | $0.982 \pm 0.017$ |
| JS to unigram model | $0.800 \pm 0.001$ | $0.802 \pm 0.001$ | $0.804 \pm 0.001$ | $0.995 \pm 0.002$ | $0.997 \pm 0.003$ |
| JS to bigram model | $0.469 \pm 0.001$ | $0.471 \pm 0.001$ | $0.462 \pm 0.001$ | $1.02 \pm 0.00$ | $1.02 \pm 0.00$ |

# C Swaps

**Table 5:** Swaps. $A$: parent's optimizer; $B$: optimizer after the swap; $s$: fork step.

| | |
|---|---|
| Parents | the d0 runs of AdamW and Muon (Table 1) |
| Warm state | the other optimizer, stepped on the same mini-batch gradients with learning rate 0: momenta, second moments and step counts evolve as if training; parameters follow $A$ alone |
| Fork steps $s$ | 50, 100, 200, 400, 700, 1000, 1500, 2500, 4000 |
| Saved at $s$ | parameters; states of both optimizers |
| Swap $A \to B$ loads | parameters at $s$; $B$'s warm state |
| Hyperparameters | $B$'s (Table 2) |
| Steps | $s$ to 10,000, on the parent's schedule and batches (data order 0) |
| Swap forks | 2 directions $\times$ 9 steps = 18 |
| Metrics (Fig. 3) | at $s + 0$, 5, 10, 25, 50, 100, 250, 500, 1000, 2000, 4000, 6000 (each with the next two steps, for the updates) and every 1000 steps |
| Same-batch control | the parent itself ($A$ continued with the same batches) |